

Unit 2. Matrix Algebra and Subspaces

1. Inverses and the inverse algorithm

An **inverse** of $A \in \mathbb{R}^{n \times n}$ is an $n \times n$ matrix B satisfying both $AB = I_n$ and $BA = I_n$. A matrix with an inverse is **invertible**. Its inverse is unique: if B, C are inverses, associativity gives

$$B = BI = B(AC) = (BA)C = IC = C.$$

Write this unique matrix as A^{-1} . Since $AA^{-1} = A^{-1}A = I_n$, A is the inverse of A^{-1} : $(A^{-1})^{-1} = A$.

Proposition 2.1 (Product and transpose inverses). If A, B are invertible square matrices of the same size, then

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A^\top)^{-1} = (A^{-1})^\top.$$

Proof. The proposed inverse satisfies both defining identities:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = I, \quad (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = I.$$

Transposing $AA^{-1} = I$ and $A^{-1}A = I$, using Unit 1, Proposition 1.3, gives

$$(A^{-1})^\top A^\top = I, \quad A^\top (A^{-1})^\top = I.$$

Thus the proposed matrices are inverses by definition. □

Repeated application of the product formula gives $(A_k \cdots A_1)^{-1} = A_1^{-1} \cdots A_k^{-1}$: append one factor and apply the two-factor formula, starting with a single factor. If A is invertible, then $AX = AY$ implies $X = Y$, by multiplying on the left by A^{-1} . Similarly $XA = YA$ permits multiplication on the right by A^{-1} . The side of multiplication matters.

Recording a row operation

An **elementary matrix** is obtained by applying one elementary row operation to I_m . It records that operation by left multiplication.

Proposition 2.2 (Elementary matrices). If E is obtained by a row operation on I_m , then EA is obtained by the same operation on every m -row matrix A . Each E is invertible, with inverse given by the reverse operation.

Proof. Row i of EA is $\sum_k e_{ik} \text{row}_k(A)$: its entry in column j is the product formula $\sum_k e_{ik} a_{kj}$. Thus the three row operations on I_m give the following rows of EA :

- Swap r, s : rows r, s become $\text{row}_s(A), \text{row}_r(A)$.
- Scale row r by $c \neq 0$: row r becomes $c \text{row}_r(A)$.
- Add cR_s to R_r , $r \neq s$: row r becomes $\text{row}_r(A) + c \text{row}_s(A)$.

All other rows are unchanged.

The reverse operations are the same swap, scaling by $1/c$, and adding $-c$ times the unchanged source row. Let F record the reverse operation. Applying either order to I_m gives $FE = EF = I_m$, so $F = E^{-1}$. □

For example, adding c times row 1 to row 3 of a three-row matrix is recorded by

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{pmatrix}, \quad E^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -c & 0 & 1 \end{pmatrix}.$$

The change $c \mapsto -c$ records the reverse row operation.

After row operations $E_1, \dots, E_k$, the result is

$$H = E_k \cdots E_1 A = EA, \quad E^{-1} = E_1^{-1} \cdots E_k^{-1}.$$

Thus E is invertible. A row operation is a reversible change in the equations; the product E records all the changes in their actual order.

Theorem 2.3 (Square invertibility criterion). For $A \in \mathbb{R}^{n \times n}$, the following are equivalent:

- (a) A is invertible.
- (b) A is row equivalent to I_n .
- (c) $Ax = 0$ has only the solution $x = 0$.
- (d) For every $b \in \mathbb{R}^n$, $Ax = b$ has exactly one solution.

Proof. • (a) $\implies$ (c): $Ax = 0$ gives $x = A^{-1}Ax = 0$.

- (c) $\implies$ (b): by Unit 1, Theorems 1.5 and 1.6, elimination can leave no free variable. There are therefore n pivots. Their strictly increasing column indices must be $1, \dots, n$, so the echelon matrix is upper triangular: all entries below the diagonal are zero. Its diagonal entries are nonzero. Divide each row by its diagonal entry. The last row is now $e_n^\top$; subtract suitable multiples of it from all earlier rows to clear the last column. The next-to-last row is then $e_{n-1}^\top$; clear its column above the diagonal. Continuing upwards produces I_n . Each operation is permitted, proving (b).
- (b) $\implies$ (a): record the reduction as $EA = I_n$, where E is a product of elementary matrices. By Proposition 2.2 and the inverse product law, E is invertible. Multiplying by E^{-1} gives $A = E^{-1}$. Hence A has inverse E .
- (a) $\implies$ (d): $A(A^{-1}b) = b$ proves existence; $Ax = b$ forces $x = A^{-1}b$, proving uniqueness.
- (d) $\implies$ (c): take $b = 0$; its unique solution is zero.

□

In particular an invertible matrix is a product of elementary matrices: if $E_k \cdots E_1 A = I$, then $A = E_1^{-1} \cdots E_k^{-1}$, and each inverse is elementary. Conversely any such product is invertible by the product law.

Proposition 2.4 (One-sided inverses for square matrices). For square A, B of the same size, either $BA = I$ or $AB = I$ implies that $B = A^{-1}$.

Proof. If $BA = I$ and $Ax = 0$, then $x = BAx = 0$. Theorem 2.3 makes A invertible, and $B = B(AA^{-1}) = (BA)A^{-1} = A^{-1}$. If $AB = I$, interchange A, B in the preceding argument. It gives $A = B^{-1}$, so $B = (B^{-1})^{-1} = A^{-1}$. □

For rectangular matrices one identity need not imply the other:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (1), \quad \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq I_2.$$

These are the standard-coordinate inclusion and selection matrices.

Computing an inverse

Apply the same row operations to both halves of $[A \mid I_n]$. If the left half becomes I_n , the result is

$$[A \mid I_n] \mapsto [EA \mid E] = [I_n \mid A^{-1}].$$

The justification is exactly the implication (b) to (a) above. If elimination of A leaves a free variable, then $N(A) \neq \{0\}$ by back substitution, so A is not invertible.

Example 2.1 (An inverse by row operations). For $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 3 & 4 & 3 \end{pmatrix}$,

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 3 & 4 & 3 & 0 & 0 & 1 \end{array} \right] &\xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 3 & 4 & 3 & 0 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \leftarrow R_3 - 3R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \end{array} \right] \\ &\xrightarrow{\substack{R_3 \leftarrow R_3 - R_2 \\ R_3 \leftarrow -R_3}} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right] \\ &\xrightarrow{\substack{R_1 \leftarrow R_1 - R_3 \\ R_2 \leftarrow R_2 - R_3}} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & -2 & 1 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right] \\ &\xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right]. \end{aligned}$$

Thus

$$A^{-1} = B = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -3 & 1 \\ 1 & 3 & -1 \end{pmatrix}.$$

Check both sides using rows:

$$\begin{aligned} AB &= \begin{pmatrix} \text{row}_2(B) + \text{row}_3(B) \\ \text{row}_1(B) + \text{row}_2(B) + \text{row}_3(B) \\ 3\text{row}_1(B) + 4\text{row}_2(B) + 3\text{row}_3(B) \end{pmatrix} = I_3, \\ BA &= \begin{pmatrix} -\text{row}_1(A) + \text{row}_2(A) \\ -3\text{row}_2(A) + \text{row}_3(A) \\ \text{row}_1(A) + 3\text{row}_2(A) - \text{row}_3(A) \end{pmatrix} = I_3. \end{aligned}$$

The full solution of $Ax = b$ is therefore $x = (-b_1 + b_2, -3b_2 + b_3, b_1 + 3b_2 - b_3)^\top$.

Proposition 2.5 (Row operations preserve column relations). If $H = EA$ with E invertible, then $N(H) = N(A)$. More explicitly, writing the columns as $h_j = Ea_j$,

$$\sum_j c_j h_j = 0 \iff \sum_j c_j a_j = 0.$$

Proof. The left sum equals $E \sum_j c_j a_j$. A zero sum on the right gives a zero sum on the left; multiplying a zero sum on the left by E^{-1} gives a zero sum on the right. Taking arbitrary coefficient columns c proves the null-space equality. $\square$

Row operations may change the columns, but preserve every coefficient relation.

2. Block multiplication and trace

Partitioned products

Partition the rows and columns into consecutive, nonempty groups. Write

$$A = [A_{ij}]_{r \times s} \in \mathbb{R}^{m \times n}, \quad B = [B_{jk}]_{s \times t} \in \mathbb{R}^{n \times p},$$

where $r \times s$ and $s \times t$ count the *blocks*, and

$$A_{ij} \in \mathbb{R}^{m_i \times n_j}, \quad B_{jk} \in \mathbb{R}^{n_j \times p_k}, \quad m = \sum_{i=1}^r m_i, \quad n = \sum_{j=1}^s n_j, \quad p = \sum_{k=1}^t p_k.$$

Thus the column groups of A match the row groups of B , with the same respective sizes $n_1, \dots, n_s$. The product has r block rows and t block columns:

$$AB = C = [C_{ik}]_{r \times t}, \quad C_{ik} = \sum_{j=1}^s A_{ij} B_{jk} \in \mathbb{R}^{m_i \times p_k}.$$

All summands have the displayed size; the blocks need not be square.

Proof. • Fix i, k and a within-block position $1 \leq \alpha \leq m_i$, $1 \leq \beta \leq p_k$. Let x be the corresponding full row of A , and y the corresponding full column of B . Put

$$N_0 = 0, \quad N_j = n_1 + \dots + n_j \quad (1 \leq j \leq s).$$

- In the j th pair of matching groups,

$$x_{N_{j-1}+\ell} = (A_{ij})_{\alpha\ell}, \quad y_{N_{j-1}+\ell} = (B_{jk})_{\ell\beta} \quad (1 \leq \ell \leq n_j).$$

- Group the defining scalar product by those same intervals:

$$\begin{aligned} (C_{ik})_{\alpha\beta} &= \sum_{h=1}^n x_h y_h = \sum_{j=1}^s \sum_{\ell=1}^{n_j} x_{N_{j-1}+\ell} y_{N_{j-1}+\ell} \\ &= \sum_{j=1}^s \sum_{\ell=1}^{n_j} (A_{ij})_{\alpha\ell} (B_{jk})_{\ell\beta} = \left(\sum_{j=1}^s A_{ij} B_{jk} \right)_{\alpha\beta}. \end{aligned}$$

Every entry agrees, proving the formula. □

For $r = s = t = 2$, the formula specializes to

$$\begin{aligned} A &= \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}, \\ AB &= \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}. \end{aligned}$$

In particular, if $A = [a_1 \ \dots \ a_n]$ and the rows of B are $b_1^\top, \dots, b_n^\top$, then

$$AB = \sum_{j=1}^n a_j b_j^\top.$$

The (i, k) entry on the right is $\sum_j a_{ij} b_{jk}$, exactly that of AB . Also $[A \ B] \begin{pmatrix} x \\ y \end{pmatrix} = Ax + By$, by splitting the column combination into the two groups.

Example 2.2 (A block system). Let $A \in \mathbb{R}^{r \times r}$, $B \in \mathbb{R}^{r \times s}$, $D \in \mathbb{R}^{s \times s}$, and suppose A, D are invertible. Block multiplication writes

$$\begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix} \iff Ax + By = u, \quad Dy = v.$$

First $y = D^{-1}v$, then $x = A^{-1}u - A^{-1}BD^{-1}v$. Thus the candidate inverse is

$$G = \begin{pmatrix} A^{-1} & -A^{-1}BD^{-1} \\ 0 & D^{-1} \end{pmatrix}.$$

Both orders give the identity:

$$\begin{aligned} \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} G &= \begin{pmatrix} I_r & -BD^{-1} + BD^{-1} \\ 0 & I_s \end{pmatrix} = I_{r+s}, \\ G \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} &= \begin{pmatrix} I_r & A^{-1}B - A^{-1}B \\ 0 & I_s \end{pmatrix} = I_{r+s}. \end{aligned}$$

Hence G is the inverse.

Trace

For $A \in \mathbb{R}^{n \times n}$, its **trace** is the sum of its **diagonal entries**:

$$\operatorname{tr} A = \sum_{i=1}^n a_{ii}.$$

A matrix is **diagonal** when all entries off its main diagonal are zero; $\operatorname{diag}(d_1, \dots, d_n)$ denotes the matrix with those diagonal entries.

Proposition 2.6 (Trace identities). For square matrices of the same size,

$$\operatorname{tr}(sA + tB) = s \operatorname{tr} A + t \operatorname{tr} B, \quad \operatorname{tr}(A^\top) = \operatorname{tr} A.$$

For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times m}$,

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Proof. The first identity is $\sum_i (sa_{ii} + tb_{ii}) = s \sum_i a_{ii} + t \sum_i b_{ii}$. Transposition leaves every diagonal entry unchanged, proving the second. For the last identity, the two square products may have different sizes, but

$$\operatorname{tr}(AB) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij} = \operatorname{tr}(BA).$$

□

If A, B, C have sizes $m \times n$, $n \times p$, $p \times m$, apply the two-factor result to A and BC , then to B and CA , to obtain

$$\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB).$$

Proposition 2.7 (A trace that detects zero). For a real $m \times n$ matrix A ,

$$\operatorname{tr}(A^\top A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2, \quad \operatorname{tr}(A^\top A) = 0 \iff A = 0.$$

Proof. The j th diagonal entry of $A^\top A$ is $\sum_i a_{ij}^2$; summing over j proves the formula. Every summand is nonnegative. Their sum is zero exactly when each entry a_{ij} is zero. □

Example 2.3 (An impossible matrix equation). Can real 3×3 matrices X, Y satisfy

$$XY - YX = \operatorname{diag}(1, 0, 0)?$$

The left side has trace $\operatorname{tr}(XY) - \operatorname{tr}(YX) = 0$; the right side has trace 1. Equality would force $0 = 1$, so no such matrices exist.

3. Linear spaces and subspaces

A **real linear space** of matrices is a nonempty collection of matrices of one fixed size, closed under addition and multiplication by real numbers. Its elements are also called vectors. The addition and scalar multiplication are the usual entrywise operations. Such a collection contains zero: choose one member X , then $0X = 0$. It contains $-X = (-1)X$ whenever it contains X . The eight coordinate laws from Unit 1 still hold; closure keeps every sum and scalar multiple in the collection.

The full collection $\mathbb{R}^{m \times n}$ is a linear space: sums and scalar multiples have the same size. This includes the spaces of columns $\mathbb{R}^m$ and of scalars $\mathbb{R}$. For real polynomials of degree at most d , including the zero polynomial, use the coefficient column

$$p(t) = a_0 + a_1t + \cdots + a_dt^d \quad \longleftrightarrow \quad (a_0, \dots, a_d)^\top.$$

Here a polynomial means its finite coefficient expression; equality means equality of coefficients. Addition and scaling are coordinatewise, so this is another notation for a linear space, denoted by $\mathcal{P}_d$.

A **subspace** of a linear space V is a subset that is itself a linear space with the inherited operations.

Proposition 2.8 (Subspace test). A subset $W \subseteq V$ is a subspace if and only if it is nonempty and

$$u, v \in W, \quad s, t \in \mathbb{R} \quad \implies \quad su + tv \in W.$$

Proof. A subspace is nonempty and is closed under each scalar multiplication and then addition, so it satisfies the displayed condition. Conversely choose $u \in W$. Taking both coefficients zero gives $0 \in W$. Taking $s = t = 1$ proves addition closure; taking $t = 0$ proves scalar closure. The coordinate laws are inherited from V , so W is a linear space. $\square$

Both $\{0\}$ and V satisfy this test. A subspace must contain zero, but this alone is insufficient: $\{0, 1\} \subseteq \mathbb{R}$ contains zero and fails addition closure because $1 + 1 = 2 \notin \{0, 1\}$.

Example 2.4 (Homogeneous conditions). In $\mathbb{R}^4$, consider

$$W = \{(a, b, c, d)^\top : a + 2b = c - d\}.$$

Define $f(a, b, c, d) = a + 2b - c + d$. Entrywise addition and scaling give

$$f(su + tv) = sf(u) + tf(v).$$

Thus $f(0) = 0$, and $f(u) = f(v) = 0$ implies $f(su + tv) = 0$. The subspace test proves that W is a subspace. Solving for a gives its complete description

$$W = \left\{ b \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + d \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix} : b, c, d \in \mathbb{R} \right\}.$$

Every vector in the displayed set satisfies the condition, and every member of W is obtained using its own last three coordinates.

In contrast, the set of $n \times n$ matrices satisfying $A^2 = I_n$ for $n \geq 1$ is not a subspace: $0^2 \neq I_n$.

Unit 1, Proposition 1.8, proves that $\mathcal{C}(A) \subseteq \mathbb{R}^m$ and $N(A) \subseteq \mathbb{R}^n$ are subspaces. The set of symmetric $n \times n$ matrices is another example: $0^\top = 0$, and for symmetric S, T , $(sS + tT)^\top = sS^\top + tT^\top = sS + tT$. For fixed $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{p \times q}$, the set $\{X \in \mathbb{R}^{n \times p} : AXB = 0\}$ is a subspace of $\mathbb{R}^{n \times p}$: it contains zero, and

$$A(sX + tY)B = sAXB + tAYB = 0$$

whenever $AXB = AYB = 0$.

Spanning sets

For a finite list $v_1, \dots, v_k \in V$, define its **span** by

$$\text{span}(v_1, \dots, v_k) = \left\{ \sum_{j=1}^k c_j v_j : c_1, \dots, c_k \in \mathbb{R} \right\}.$$

The span of the empty list is $\{0\}$. A list **spans** W when this set equals W ; its members are then **generators** of W . Thus $\mathcal{C}(A)$ is the span of the columns of A .

Proposition 2.9 (The smallest subspace containing a list). The span of $v_1, \dots, v_k$ is a subspace containing every v_j . It is contained in every subspace containing the list.

Proof. The all-zero coefficients give zero. If $x = \sum_j a_j v_j$ and $y = \sum_j b_j v_j$, then $sx + ty = \sum_j (sa_j + tb_j)v_j$, proving closure. Putting coefficient 1 on v_j and 0 on the others gives v_j itself. Finally a subspace containing all the v_j contains all their finite linear combinations, by repeated addition and scaling. For an empty list its span is $\{0\}$, which is in every subspace. $\square$

Proposition 2.10 (Removing a redundant generator). If v_j is a combination of the other members of a generating list, removing it does not change the span.

Proof. Write $v_j = \sum_{i \neq j} d_i v_i$. Every combination of the full list is

$$\sum_i c_i v_i = \sum_{i \neq j} (c_i + c_j d_i) v_i,$$

so it is in the shorter span. Conversely a combination of the shorter list is a combination of the original list with coefficient zero on v_j . These give the two inclusions. $\square$

Example 2.5 (Matrices commuting with a fixed matrix). Find all matrices B commuting with $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Writing $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, multiplication gives

$$AB = \begin{pmatrix} a + 2c & b + 2d \\ 3a + 4c & 3b + 4d \end{pmatrix}, \quad BA = \begin{pmatrix} a + 3b & 2a + 4b \\ c + 3d & 2c + 4d \end{pmatrix}.$$

Equality is equivalent to

$$2c = 3b, \quad 2d = 2a + 3b, \quad c = d - a.$$

The first two give $c = 3b/2$, $d = a + 3b/2$, which also satisfy the third. Set $\alpha = a$, $\beta = b/2$. All solutions are exactly

$$B = \begin{pmatrix} \alpha & 2\beta \\ 3\beta & \alpha + 3\beta \end{pmatrix} = \alpha I_2 + \beta \begin{pmatrix} 0 & 2 \\ 3 & 3 \end{pmatrix}.$$

Conversely these entries satisfy all three equations, hence $AB = BA$. The solution set is the span of the two displayed matrices and is therefore a subspace. A direct check also works: $A0 = 0A$, and if $AB = BA$, $AC = CA$, then

$$A(sB + tC) = sAB + tAC = sBA + tCA = (sB + tC)A.$$

4. Intersections, sums, and translates

Proposition 2.11 (Intersection and sum). If U, W are subspaces of the same V , then so are

$$U \cap W, \quad U + W := \{u + w : u \in U, w \in W\}.$$

Moreover, $U + W$ is the smallest subspace containing both U and W .

Proof. Zero belongs to both sets. If $x, y \in U \cap W$, then $sx + ty$ belongs to U and to W , hence to the intersection. For $x = u_1 + w_1$, $y = u_2 + w_2$,

$$sx + ty = (su_1 + tw_1) + (tu_2 + tw_2) \in U + W.$$

Thus the sum is also a subspace. Taking $w = 0$ or $u = 0$ shows that it contains U and W . Any subspace containing both must contain all their sums, giving the asserted minimality. $\square$

Proposition 2.12 (When is a union a subspace?). For subspaces U, W of V , the union $U \cup W$ is a subspace if and only if $U \subseteq W$ or $W \subseteq U$.

Proof. • If either inclusion holds, the union is the larger subspace.

- If neither holds, choose $u \in U \setminus W$, $w \in W \setminus U$. Then

$$u + w \in U \implies w = (u + w) - u \in U, \quad u + w \in W \implies u = (u + w) - w \in W.$$

Both contradict the choices. Hence $u, w \in U \cup W$, but $u + w \notin U \cup W$: addition closure fails. $\square$

Proposition 2.13 (Uniqueness of a sum decomposition). Let U, W be subspaces of the same real linear space V . Every $x \in U + W$ has a unique representation $x = u + w$ with $u \in U$, $w \in W$, if and only if $U \cap W = \{0\}$. When this holds the sum is called a **direct sum**, written $U \oplus W$.

Proof. Existence follows from the definition of $U + W$. If $u + w = u' + w'$, then $u - u' = w' - w \in U \cap W$. A zero intersection forces $u = u'$, $w = w'$. Conversely, for $z \in U \cap W$, the two representations $z = z + 0 = 0 + z$ must agree term by term. Hence $z = 0$. $\square$

Example 2.6 (Symmetric and skew-symmetric parts). A matrix K is **skew-symmetric** if $K^\top = -K$. The symmetric and skew-symmetric $n \times n$ matrices each form a subspace: transpose linearity proves closure, and zero belongs to both. Every $A \in \mathbb{R}^{n \times n}$ can be written

$$A = S + K, \quad S = \frac{A + A^\top}{2}, \quad K = \frac{A - A^\top}{2}.$$

Indeed their sum is A , while the transpose laws give $S^\top = S$, $K^\top = -K$. A matrix in both subspaces satisfies $X = X^\top = -X$, so $2X = 0$ and every entry is zero. Proposition 2.13 therefore proves that this decomposition is unique.

For $p \in V$ and a subspace W , the set $p + W = \{p + w : w \in W\}$ is a **translate** of W , also called an **affine subspace**.

Proposition 2.14 (A translate that is a subspace). The set $p + W$ is a subspace if and only if $p \in W$; in that case $p + W = W$.

Proof. If $p \in W$, then $p + w \in W$ for all $w \in W$. Conversely every $w \in W$ equals $p + (w - p)$, with $w - p \in W$, proving equality. If $p + W$ is a subspace, it contains zero, so $p + w = 0$ for some $w \in W$. Hence $p = -w \in W$. $\square$

For a consistent system $Ax = b$, Unit 1, Theorem 1.9, gives the affine subspace $x_0 + N(A)$. It is a linear subspace exactly when $b = 0$: if it is a subspace then zero solves the equation, forcing $b = A0 = 0$; if $b = 0$ its solution set is the subspace $N(A)$.

For fixed $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{p \times q}$, $C \in \mathbb{R}^{m \times q}$, suppose $X_0 \in \mathbb{R}^{n \times p}$ satisfies $AX_0B = C$. Then, with $X, Z \in \mathbb{R}^{n \times p}$,

$$\{X : AXB = C\} = X_0 + \{Z : AZB = 0\}.$$

Indeed $AXB = C$ implies $A(X - X_0)B = 0$, and conversely $AZB = 0$ implies $A(X_0 + Z)B = C$.

Exercises and solutions

Suggested practice: Exercises 2.1, 2.3, 2.7, 2.14, 2.16, and 2.18.

Inverses and matrix equations

Exercise 2.1 (Two inverse calculations). Find the inverses of

$$(a) \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc \neq 0; \quad (b) \quad \begin{pmatrix} 2 & -1 & 0 \\ 2 & 1 & -1 \\ 1 & 0 & 4 \end{pmatrix}.$$

Solution. (a) Put $\delta = ad - bc$. Multiplication in either order gives

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \delta I_2.$$

Thus the inverse is $\delta^{-1} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Division by δ is valid because $\delta \neq 0$.

(b) Solve $Ax = y$ for an arbitrary $y = (y_1, y_2, y_3)^\top$. The first two equations give

$$x_2 = 2x_1 - y_1, \quad x_3 = 2x_1 + x_2 - y_2 = 4x_1 - y_1 - y_2.$$

The third becomes $17x_1 = y_3 + 4y_1 + 4y_2$. Substitution yields

$$x = \frac{1}{17} \begin{pmatrix} 4 & 4 & 1 \\ -9 & 8 & 2 \\ -1 & -1 & 4 \end{pmatrix} y.$$

Writing the displayed integer matrix as K , direct multiplication gives $AK = KA = 17I_3$. Therefore $A^{-1} = K/17$.

Exercise 2.2 (Inverse identities). Let A be invertible. Prove the following statements.

- (a) If $\lambda \neq 0$, then $(\lambda A)^{-1} = \lambda^{-1} A^{-1}$.
- (b) $(A^{-1})^{-1} = A$.
- (c) If $AB = BA$ and B is invertible, then $AB^{-1} = B^{-1}A$.

Solution. (a) Both $(\lambda A)(\lambda^{-1} A^{-1})$ and $(\lambda^{-1} A^{-1})(\lambda A)$ equal I .

(b) The equations $AA^{-1} = A^{-1}A = I$ also say that A is the inverse of A^{-1} .

(c) Multiply $AB = BA$ on the left and on the right by B^{-1} :

$$B^{-1}ABB^{-1} = B^{-1}BAB^{-1}, \quad B^{-1}A = AB^{-1}.$$

Exercise 2.3 (One right-hand side is enough). Let A be $n \times n$ and fix $b \in \mathbb{R}^n$. Prove that A is invertible if and only if $Ax = b$ has exactly one solution.

Solution. If A is invertible, $A^{-1}b$ is a solution and multiplication by A^{-1} shows that every solution equals it. Conversely, let x_0 be the unique solution. If $Az = 0$, then $A(x_0 + z) = b$, so uniqueness gives $z = 0$. Thus the homogeneous system has only the zero solution, and the square inverse criterion (Theorem 2.3) gives the inverse. Concretely, elimination can have no free variable; all n columns have pivots, and scaling and upward elimination reduce A to I_n .

Exercise 2.4 (Rectangular left and right inverses). Let A be $m \times n$.

- (a) If $AR = I_m$ for some R , prove that $n \geq m$.
 (b) If $LA = I_n$ for some L , prove that $m \geq n$.

Solution. (a) If $m > n$, the system $A^\top y = 0$ has n equations in m unknowns, so Unit 1, Proposition 1.7, gives a nonzero solution y . Transposing gives $y^\top A = 0$, hence

$$y^\top = y^\top I_m = y^\top AR = 0,$$

a contradiction. (b) If $n > m$, elimination gives a nonzero z with $Az = 0$. Then $z = I_n z = LAz = 0$, again a contradiction.

Exercise 2.5 (Matrices that are their own inverses). A square matrix A is called *involutory* if $A^2 = I$.

- (a) Prove that A is involutory if and only if $(I - A)(I + A) = 0$.
 (b) Prove that $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is involutory if and only if either

$$d = -a, \quad a^2 + bc = 1, \quad \text{or} \quad b = c = 0, \quad d = a = 1 \text{ or } d = a = -1.$$

Solution. (a) Distributivity gives $(I - A)(I + A) = I - A^2$. (b) The equation $A^2 = I_2$ is precisely

$$a^2 + bc = 1, \quad b(a + d) = 0, \quad c(a + d) = 0, \quad d^2 + bc = 1.$$

If $a + d = 0$, these reduce to $d = -a$ and $a^2 + bc = 1$. If $a + d \neq 0$, the middle equations force $b = c = 0$. Then $a^2 = d^2 = 1$; the condition $a + d \neq 0$ forces $a = d = 1$ or $a = d = -1$. Conversely, either displayed set of conditions satisfies all four equations, so both families are involutory.

Exercise 2.6 (Changing an invertible matrix by an outer product). Let A be invertible and let a, b be columns of the appropriate size. Prove the following formulas under their stated scalar conditions:

- (a) If $a^\top A^{-1}a \neq -1$, then

$$(A + aa^\top)^{-1} = A^{-1} - \frac{A^{-1}aa^\top A^{-1}}{1 + a^\top A^{-1}a}.$$

- (b) If $a^\top A^{-1}a \neq 1$, then

$$(A - aa^\top)^{-1} = A^{-1} + \frac{A^{-1}aa^\top A^{-1}}{1 - a^\top A^{-1}a}.$$

- (c) If $b^\top A^{-1}a \neq -1$, then

$$(A + ab^\top)^{-1} = A^{-1} - \frac{A^{-1}ab^\top A^{-1}}{1 + b^\top A^{-1}a}.$$

Solution. Prove (c) first. Put $t = b^\top A^{-1}a$ and $X = A^{-1} - A^{-1}ab^\top A^{-1}/(1 + t)$. The denominator is nonzero. Associativity, with t a scalar, gives

$$\begin{aligned} (A + ab^\top)X &= I + ab^\top A^{-1} - \frac{ab^\top A^{-1} + atb^\top A^{-1}}{1 + t} = I, \\ X(A + ab^\top) &= I + A^{-1}ab^\top - \frac{A^{-1}ab^\top + A^{-1}atb^\top}{1 + t} = I. \end{aligned}$$

This proves (c). Setting $b = a$ proves (a). Setting $b = -a$ gives $t = -a^\top A^{-1}a$ and changes the numerator's sign, which gives exactly (b).

Blocks and column operations

Exercise 2.7 (Compatible block calculations). Use the indicated partitions to find $A + B$, AC , and $A^\top$, where

$$A = \left(\begin{array}{ccc|cc} 1 & 3 & -2 & 1 & 2 \\ 6 & 8 & 0 & -1 & 6 \\ 0 & 0 & 1 & 4 & 1 \end{array} \right), \quad B = \left(\begin{array}{ccc|cc} 1 & -3 & -2 & 4 & 1 \\ 6 & 2 & 6 & 2 & 0 \\ 1 & 0 & 2 & 0 & 1 \end{array} \right),$$

$$C = \left(\begin{array}{cc|cc} 1 & 0 & 5 & 1 \\ 0 & 2 & 0 & 0 \\ -1 & 0 & 3 & 1 \\ \hline 3 & 5 & 0 & 2 \\ 2 & -1 & 3 & 1 \end{array} \right).$$

Solution. Addition is entrywise within corresponding blocks:

$$A + B = \left(\begin{array}{ccc|cc} 2 & 0 & -4 & 5 & 3 \\ 12 & 10 & 6 & 1 & 6 \\ 1 & 0 & 3 & 4 & 2 \end{array} \right).$$

Writing the blocks as A_{ij}, C_{ij} , the four blocks of AC are

$$\begin{aligned} A_{11}C_{11} + A_{12}C_{21} &= \begin{pmatrix} 3 & 6 \\ 6 & 16 \end{pmatrix} + \begin{pmatrix} 7 & 3 \\ 9 & -11 \end{pmatrix} = \begin{pmatrix} 10 & 9 \\ 15 & 5 \end{pmatrix}, \\ A_{11}C_{12} + A_{12}C_{22} &= \begin{pmatrix} -1 & -1 \\ 30 & 6 \end{pmatrix} + \begin{pmatrix} 6 & 4 \\ 18 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 48 & 10 \end{pmatrix}, \\ A_{21}C_{11} + A_{22}C_{21} &= \begin{pmatrix} -1 & 0 \end{pmatrix} + \begin{pmatrix} 14 & 19 \end{pmatrix} = \begin{pmatrix} 13 & 19 \end{pmatrix}, \\ A_{21}C_{12} + A_{22}C_{22} &= \begin{pmatrix} 3 & 1 \end{pmatrix} + \begin{pmatrix} 3 & 9 \end{pmatrix} = \begin{pmatrix} 6 & 10 \end{pmatrix}. \end{aligned}$$

Consequently,

$$AC = \left(\begin{array}{cc|cc} 10 & 9 & 5 & 3 \\ 15 & 5 & 48 & 10 \\ \hline 13 & 19 & 6 & 10 \end{array} \right), \quad A^\top = \left(\begin{array}{cc|cc} 1 & 6 & 0 \\ 3 & 8 & 0 \\ -2 & 0 & 1 \\ \hline 1 & -1 & 4 \\ 2 & 6 & 1 \end{array} \right).$$

Transposition interchanges the block positions and transposes each block.

Exercise 2.8 (Reading an inverse in blocks). Suppose $A = [A_1 \ A_2]$ is invertible, where A_1 has p columns and A_2 has q columns. Partition

$$A^{-1} = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix}, \quad B_1 : p \times (p+q), \quad B_2 : q \times (p+q).$$

Prove that

$$B_1A_1 = I_p, \quad B_1A_2 = 0, \quad B_2A_1 = 0, \quad B_2A_2 = I_q,$$

and that $A_1B_1 = I_{p+q} - A_2B_2$ and $A_2B_2 = I_{p+q} - A_1B_1$.

Solution. The two defining inverse identities give

$$A^{-1}A = \begin{pmatrix} B_1A_1 & B_1A_2 \\ B_2A_1 & B_2A_2 \end{pmatrix} = \begin{pmatrix} I_p & 0 \\ 0 & I_q \end{pmatrix}, \quad AA^{-1} = A_1B_1 + A_2B_2 = I_{p+q}.$$

Comparing the four blocks proves the first four equations; rearranging the last equation proves each of the last two.

Exercise 2.9 (A lower triangular block inverse). Let C be $n \times m$.

(a) Find the inverse of $\begin{pmatrix} I_m & 0 \\ C & I_n \end{pmatrix}$.

(b) More generally, if A and D are invertible of orders m and n , find the inverse of $L = \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$.

Solution. For (b), try $X = \begin{pmatrix} A^{-1} & 0 \\ Q & D^{-1} \end{pmatrix}$. The lower-left block of LX is $CA^{-1} + DQ$, so this block vanishes exactly when $Q = -D^{-1}CA^{-1}$. For this choice,

$$LX = \begin{pmatrix} I_m & 0 \\ CA^{-1} - DD^{-1}CA^{-1} & I_n \end{pmatrix} = I_{m+n},$$

$$XL = \begin{pmatrix} I_m & 0 \\ -D^{-1}CA^{-1}A + D^{-1}C & I_n \end{pmatrix} = I_{m+n}.$$

Thus $L^{-1} = \begin{pmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{pmatrix}$. Taking $A = I_m, D = I_n$ proves (a): replace C by $-C$.

Exercise 2.10 (Column operations as right multiplication). (a) Prove that an elementary column operation on an $m \times n$ matrix A can be performed by right multiplication by an $n \times n$ elementary matrix.

(b) For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$, give the right multiplier and the resulting matrix for each separate operation:
swap the two columns; multiply column 1 by 3; add 5 times column 1 to column 2.

Solution. (a) Write $A = [a_1 \cdots a_n]$. The j th column of AF is $\sum_k f_{kj}a_k$. Swapping columns i, j of I_n therefore swaps columns i, j of A ; the resulting F is also obtained by swapping those rows of I_n . Scaling column j of I_n by a nonzero c gives a diagonal elementary matrix and scales a_j by c . Finally, adding c times column i to column j of I_n inserts c in position (i, j) . This same matrix is obtained by adding c times row j to row i of I_n , and its right action replaces a_j by $a_j + ca_i$. Thus each multiplier is elementary.

(b) In the order requested,

$$\begin{array}{cc} F & AF \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{pmatrix} \\ \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 3 & 2 \\ 9 & 4 \\ 15 & 6 \end{pmatrix} \\ \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 7 \\ 3 & 19 \\ 5 & 31 \end{pmatrix}. \end{array}$$

Trace and matrix identities

Exercise 2.11 (What cyclic trace permits). (a) For $m \times n$ matrices A, B , prove that $\text{tr}(A^\top B) = \text{tr}(BA^\top) = \text{tr}(AB^\top) = \text{tr}(B^\top A)$.

(b) Prove that $\text{tr}(Aaa^\top) = a^\top Aa$ for square A and a compatible column a .

(c) Prove that $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$, specifying the compatible sizes.

(d) Must $\text{tr}(ABC) = \text{tr}(ACB)$ also hold?

Solution. (a) Each of the four expressions equals $\sum_{i=1}^m \sum_{j=1}^n a_{ij}b_{ij}$: for example,

$$\text{tr}(A^\top B) = \sum_j \sum_i a_{ij}b_{ij}, \quad \text{tr}(AB^\top) = \sum_i \sum_j a_{ij}b_{ij}.$$

The other two reverse the order of the scalar factors. (b) Applying the rectangular cyclic identity to $(Aa)a^\top$ gives $\text{tr}((Aa)a^\top) = \text{tr}(a^\top Aa) = a^\top Aa$, since the latter matrix is 1×1 . (c) Take $A : m \times n$, $B : n \times p$, and $C : p \times m$. Applying the two-factor cyclic identity to $(AB)C$ and $(BC)A$ gives

$$\text{tr}(ABC) = \text{tr}(CAB), \quad \text{tr}(BCA) = \text{tr}(ABC).$$

(d) No. For example, when $A : 1 \times 2$, $B : 2 \times 3$, and $C : 3 \times 1$, ABC is square, whereas ACB is not even defined: AC has incompatible sizes.

Exercise 2.12 (Interchanging symmetric factors). (a) If A, B, C are symmetric matrices of the same order, prove $\text{tr}(ABC) = \text{tr}(BAC)$.

(b) Show that this equality can fail without symmetry.

Solution. (a) Transposing the product and then moving its first factor to the end gives

$$\text{tr}(ABC) = \text{tr}((ABC)^\top) = \text{tr}(CBA) = \text{tr}(BAC).$$

(b) Take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Here $AB = 0$, so $\text{tr}(ABC) = 0$, whereas $BAC = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ has trace 2. For any larger order one may append zero rows and columns; order 1 cannot give a counterexample because real scalar multiplication commutes.

Exercise 2.13 (Testing a matrix by traces). Let A be $n \times n$. If $\text{tr}(AB) = 0$ for every $n \times n$ matrix B , must A be zero?

Solution. Yes. The hypothesis permits $B = A^\top$, so

$$0 = \text{tr}(AA^\top) = \sum_{i=1}^n \sum_{j=1}^n a_{ij}^2.$$

Each term is nonnegative, and a positive term would make the sum positive. Thus every $a_{ij} = 0$ and $A = 0$.

Exercise 2.14 (Five tests for symmetry). For a square real matrix A , prove that the following conditions are equivalent:

- (i) $A = A^\top$; (ii) $A^2 = AA^\top$;
- (iii) $\operatorname{tr}(A^2) = \operatorname{tr}(AA^\top)$; (iv) $A^2 = A^\top A$;
- (v) $\operatorname{tr}(A^2) = \operatorname{tr}(A^\top A)$.

Solution.

- Substitution of $A = A^\top$ and then taking traces give

$$(i) \implies (ii) \implies (iii), \quad (i) \implies (iv) \implies (v).$$

- The cyclic identity $\operatorname{tr}(AA^\top) = \operatorname{tr}(A^\top A)$ gives $(iii) \iff (v)$.
- To prove $(iii) \implies (i)$, put $D = A^\top - A$. Then

$$\begin{aligned} \operatorname{tr}(D^\top D) &= \operatorname{tr}(AA^\top - A^2 - (A^\top)^2 + A^\top A) \\ &= 2\operatorname{tr}(AA^\top) - 2\operatorname{tr}(A^2) = 0. \end{aligned}$$

We used $(A^\top)^2 = (A^2)^\top$ and equality of a matrix's trace and its transpose's trace. Since $\operatorname{tr}(D^\top D) = \sum_{i,j} d_{ij}^2$, all entries of D vanish. Hence $A = A^\top$.

The two chains are now linked in both directions.

Exercise 2.15 (Diagonal and off-diagonal sums). Let $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^n$ and let $D = \operatorname{diag}(a_{11}, \dots, a_{nn})$ for an $n \times n$ matrix A . Prove that

$$\mathbf{1}^\top A \mathbf{1} = \mathbf{1}^\top D \mathbf{1} + \operatorname{tr}((\mathbf{1}\mathbf{1}^\top - I_n)A).$$

Solution. The cyclic identity and trace linearity give

$$\operatorname{tr}((\mathbf{1}\mathbf{1}^\top - I_n)A) = \operatorname{tr}(\mathbf{1}^\top A \mathbf{1}) - \operatorname{tr}(A) = \mathbf{1}^\top A \mathbf{1} - \sum_i a_{ii}.$$

Also $\mathbf{1}^\top D \mathbf{1} = \sum_i a_{ii}$. Substitution proves the formula: the sum of all entries splits into diagonal and off-diagonal sums.

Subspaces and representations

Exercise 2.16 (A condition on four matrix entries). Is the set of all 3×3 real matrices A satisfying $a_{11} + a_{13} = a_{22} + a_{31}$ a subspace of $\mathbb{R}^{3 \times 3}$?

Solution. Yes. Put $g(A) = a_{11} + a_{13} - a_{22} - a_{31}$. Then $g(0) = 0$ and, by entrywise addition and scaling,

$$g(\alpha A + \beta B) = \alpha g(A) + \beta g(B).$$

Consequently $g(A) = g(B) = 0$ implies $g(\alpha A + \beta B) = 0$, which verifies both nonemptiness and closure under linear combinations.

Exercise 2.17 (Matrix subspaces). A matrix is upper triangular when $a_{ij} = 0$ for $i > j$, and lower triangular when $a_{ij} = 0$ for $i < j$. Determine whether each set is a subspace of the corresponding real matrix space:

- (a) All upper triangular $n \times n$ matrices.
- (b) All nonsymmetric $n \times n$ matrices.
- (c) All lower triangular 3×3 matrices.
- (d) All symmetric 3×3 matrices.

Solution. (a) Yes. The entries below the diagonal are zero, and $(\alpha A + \beta B)_{ij} = \alpha 0 + \beta 0 = 0$ for $i > j$. The zero matrix also has this property. (b) No. The zero matrix is symmetric and is excluded. This also settles $n = 1$, when the proposed set is empty. (c) Yes, by the same entry calculation as (a), now for $i < j$. (d) Yes. The zero matrix is symmetric, and for symmetric A, B ,

$$(\alpha A + \beta B)^\top = \alpha A^\top + \beta B^\top = \alpha A + \beta B.$$

Exercise 2.18 (All subspaces of $\mathbb{R}^2$). (a) Show that $\{(0, t)^\top : t \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.

- (b) Is $\{(1, t)^\top : t \in \mathbb{R}\}$ a subspace?
- (c) Is $\{(x, y)^\top : x \geq 0, y \geq 0\}$ a subspace?
- (d) Prove that every subspace of $\mathbb{R}^2$ is $\{0\}$, all of $\mathbb{R}^2$, or $\{ta : t \in \mathbb{R}\}$ for some nonzero $a \in \mathbb{R}^2$.

Solution. (a) The zero column belongs, and $\alpha(0, s)^\top + \beta(0, t)^\top = (0, \alpha s + \beta t)^\top$ belongs. (b) No: it excludes zero. (c) No: it contains $(1, 1)^\top$ but not its negative.

(d) Let W be a nonzero subspace, and choose $a = (a_1, a_2)^\top \neq 0$ in W . Closure gives $\{ta : t \in \mathbb{R}\} \subseteq W$. If equality holds, we have the third listed type. Otherwise take $b = (b_1, b_2)^\top \in W$ that is not a multiple of a . We claim $\delta = a_1 b_2 - a_2 b_1 \neq 0$. If $\delta = 0$ and $a_1 \neq 0$, then $b = (b_1/a_1)a$. If $\delta = 0$ and $a_1 = 0$, then $a_2 \neq 0$, the equation forces $b_1 = 0$, and $b = (b_2/a_2)a$. Both contradict the choice of b . For any $x = (x_1, x_2)^\top \in \mathbb{R}^2$, set

$$\alpha = \frac{b_2 x_1 - b_1 x_2}{\delta}, \quad \beta = \frac{a_1 x_2 - a_2 x_1}{\delta}.$$

Substitution gives $\alpha a + \beta b = x$, so $x \in W$ and $W = \mathbb{R}^2$. Finally, $\{0\}$ and $\mathbb{R}^2$ are subspaces by the subspace test. The third type contains $0 = 0a$ and is closed because $\alpha(sa) + \beta(ta) = (\alpha s + \beta t)a$.

Exercise 2.19 (When decomposition is unique). Let U, W be subspaces of one real linear space.

- (a) If $U \cap W = \{0\}$, prove that every $x \in U + W$ has a unique representation $x = u + w$ with $u \in U, w \in W$.
- (b) If $U \cap W \neq \{0\}$, prove that some $x \in U + W$ has more than one such representation.

Solution. (a) Existence is the definition of $U + W$. If $u + w = \tilde{u} + \tilde{w}$, then

$$u - \tilde{u} = \tilde{w} - w \in U \cap W = \{0\}.$$

Hence $u = \tilde{u}$ and $w = \tilde{w}$. (b) Take $z \neq 0$ in $U \cap W$. The two expressions $z = z + 0$ and $z = 0 + z$ have different U -components and different W -components. Both are admissible.

Exercise 2.20 (Adding and removing generators). Let $A_1, \dots, A_p, B_1, \dots, B_q$ belong to a linear space V of real matrices.

- (a) If $A_1, \dots, A_p$ span V , prove that the larger list also spans V .
- (b) If the larger list spans V and each B_j is a linear combination of $A_1, \dots, A_p$, prove that $A_1, \dots, A_p$ span V .

Solution. (a) Any $X \in V$ is $\sum_i x_i A_i$; append zero coefficients for the B_j . All combinations of the larger list remain in V by its closure, so its span is exactly V . (b) Write $B_j = \sum_i k_{ij} A_i$. For $X \in V$, a representation using the larger list gives

$$X = \sum_i x_i A_i + \sum_j y_j B_j = \sum_i \left(x_i + \sum_j y_j k_{ij} \right) A_i.$$

Thus every $X \in V$ belongs to the span of the A_i ; their combinations also remain in V .

Exercise 2.21 (A redundant representation). Suppose $A_1, \dots, A_k$ span a real matrix space V , and that

$$\sum_{i=1}^k z_i A_i = 0 \quad \text{for some scalars } z_1, \dots, z_k \text{ not all zero.}$$

Prove that every $A \in V$ has more than one representation as a linear combination of $A_1, \dots, A_k$.

Solution. Choose $A = \sum_i x_i A_i$, using the spanning assumption. Then

$$A = \sum_i x_i A_i + \sum_i z_i A_i = \sum_i (x_i + z_i) A_i.$$

At least one coefficient changes because at least one z_i is nonzero. Thus these are two different coefficient lists for the same matrix A .

Exercise 2.22 (A homogeneous matrix equation). Let A be 9×7 and B be 4×3 . Prove that there is a nonzero 7×4 matrix X such that $AXB = 0$.

Solution. The entries of X are 28 scalar unknowns. For each $1 \leq i \leq 9$ and $1 \leq j \leq 3$, the entry equation is

$$(AXB)_{ij} = \sum_{r=1}^7 \sum_{s=1}^4 a_{ir} b_{sj} x_{rs} = 0.$$

These are 27 homogeneous linear equations in those 28 unknowns. Elimination has at most 27 pivot variables, so at least one variable is free. Set one free variable to 1, all other free variables to 0, and solve backwards for the pivot variables. The resulting solution is nonzero and gives the required X .

Further problems on products and powers

- Exercise 2.23** (All square-zero matrices of order two). (a) Prove that a symmetric real 2×2 matrix satisfying $A^2 = 0$ must be zero.
- (b) Prove that all real 2×2 solutions of $A^2 = 0$ are precisely $A = pq^\top$, where $p, q \in \mathbb{R}^2$ and $p^\top q = 0$. Show that there are infinitely many solutions.

Solution. Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Its square is zero exactly when

$$a^2 + bc = 0, \quad b(a + d) = 0, \quad c(a + d) = 0, \quad d^2 + bc = 0.$$

If $a + d \neq 0$, the middle equations give $b = c = 0$, and the other two give $a = d = 0$, a contradiction. Thus the conditions reduce to

$$d = -a, \quad a^2 + bc = 0.$$

- (a) Symmetry gives $c = b$, so $a^2 + b^2 = 0$ and all four entries vanish.
- (b) If $a = 0$, then $bc = 0$, and the matrix has one of the forms

$$\begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & b \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ c & 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} c & 0 \end{pmatrix}.$$

These include the zero matrix. If $a \neq 0$, then $bc = -a^2 \neq 0$, and

$$A = \begin{pmatrix} a & b \\ -a^2/b & -a \end{pmatrix} = \begin{pmatrix} 1 \\ -a/b \end{pmatrix} \begin{pmatrix} a & b \end{pmatrix}.$$

In each factorization $A = pq^\top$ and $q^\top p = p^\top q = 0$. Conversely, $pq^\top pq^\top = p(q^\top p)q^\top = 0$ whenever this scalar is zero. Finally, the first displayed family has one distinct solution for every real b .

- Exercise 2.24** (Powers and recurrences). (a) Set $F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and define $x_{-1} = 0$, $x_0 = 1$, $x_n = x_{n-1} + x_{n-2}$ for $n \geq 1$. Prove that

$$F^n = \begin{pmatrix} x_n & x_{n-1} \\ x_{n-1} & x_{n-2} \end{pmatrix} \quad (n \geq 1).$$

- (b) Put $A = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$. Show that $B = A^2$ and $B^2 = -A$; compute $A^2, \dots, A^6$; and deduce that $A^6 = I$ and $B^3 = I$.
- (c) Explain the relation between this A and the recurrence $u_n = u_{n-1} - u_{n-2}$.

Solution. (a) For $n = 1$, $x_1 = 1$, and the displayed matrix is F . If the formula holds at n , multiplication on the right by F gives

$$F^{n+1} = \begin{pmatrix} x_n + x_{n-1} & x_n \\ x_{n-1} + x_{n-2} & x_{n-1} \end{pmatrix} = \begin{pmatrix} x_{n+1} & x_n \\ x_n & x_{n-1} \end{pmatrix}.$$

This proves the formula by induction. (b) Direct products give

$$A^2 = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} = B, \quad A^3 = BA = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I.$$

Multiplying successively by A now gives

$$A^4 = -A = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \quad A^5 = -B = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \quad A^6 = I.$$

Consequently $B^2 = A^4 = -A$ and $B^3 = A^6 = I$. (c) Given initial values u_0, u_{-1} , set $z_n = (u_n, u_{n-1})^\top$. Then, for $n \geq 1$,

$$z_n = Az_{n-1} \iff \begin{pmatrix} u_n \\ u_{n-1} \end{pmatrix} = \begin{pmatrix} u_{n-1} - u_{n-2} \\ u_{n-1} \end{pmatrix}.$$

Iterating this equality gives $z_n = A^n z_0$, and conversely these matrix powers satisfy the recurrence by $A^{n+1} = AA^n$.

Exercise 2.25 (A binomial expansion without commutativity). Let A, B be $n \times n$ and let p be a positive integer. For $0 \leq j \leq p$, let $\binom{p}{j}$ denote the number of j -element subsets of $\{1, \dots, p\}$.

- (a) Show that $(A + B)^p = A^p(I + A^{-1}B)^p$ need not hold, even when A is invertible.
- (b) Let $W_{p,j}$ be the sum of all products of length p containing j factors B and $p - j$ factors A , in all possible orders. Count each order separately, even when different orders give the same matrix. For example,

$$W_{3,1} = A^2B + ABA + BA^2,$$

$$W_{4,2} = A^2B^2 + ABAB + AB^2A + BA^2B + BABA + B^2A^2.$$

Prove that $(A + B)^p = \sum_{j=0}^p W_{p,j}$.

- (c) Prove that if $AB = BA$, then $(A + B)^p = \sum_{j=0}^p \binom{p}{j} A^{p-j} B^j$.
- (d) Prove that the converse in (c) holds when $p = 2$, but can fail for another positive integer p .

Solution. (a) Take $p = 2$ and $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $A^{-1}B = B$, $B^2 = 0$, and

$$(A + B)^2 = \begin{pmatrix} 1 & 3 \\ 0 & 4 \end{pmatrix}, \quad A^2(I + A^{-1}B)^2 = A^2(I + 2B) = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}.$$

Without invertibility, the proposed right side need not even be defined.

(b) At $p = 1$ the sum is $A + B$. Suppose the expansion holds at p . Multiplication on the left by $A + B$ places either A or B in front of each product of length p . Every order of $p + 1$ factors occurs exactly once: its first factor specifies the choice, and deleting it identifies the remaining order. More explicitly, setting $W_{p,-1} = W_{p,p+1} = 0$ gives

$$W_{p+1,j} = AW_{p,j} + BW_{p,j-1} \quad (0 \leq j \leq p + 1).$$

Distributing $(A + B) \sum_{j=0}^p W_{p,j}$ and grouping by the number of B factors therefore proves the expansion at $p + 1$.

(c) Under $AB = BA$, adjacent unlike factors can be interchanged, so moving every A to the left turns each summand of $W_{p,j}$ into $A^{p-j}B^j$. There are $\binom{p}{j}$ such summands, one for each choice of the j positions occupied by B . Thus $W_{p,j} = \binom{p}{j} A^{p-j}B^j$, and (b) gives the formula.

(d) For $p = 2$, comparison of

$$(A + B)^2 = A^2 + AB + BA + B^2 \quad \text{and} \quad A^2 + 2AB + B^2$$

forces $BA = AB$. For $p = 3$, take

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Multiplication gives

$$A^2 = 0, \quad AB = 0, \quad BA = B^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0, \quad B^3 = 0.$$

Also

$$A + B = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A + B)^2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (A + B)^3 = 0.$$

Since $A^3 = A^2B = AB^2 = B^3 = 0$, the claimed binomial expression for $p = 3$ also equals zero, although $AB \neq BA$.